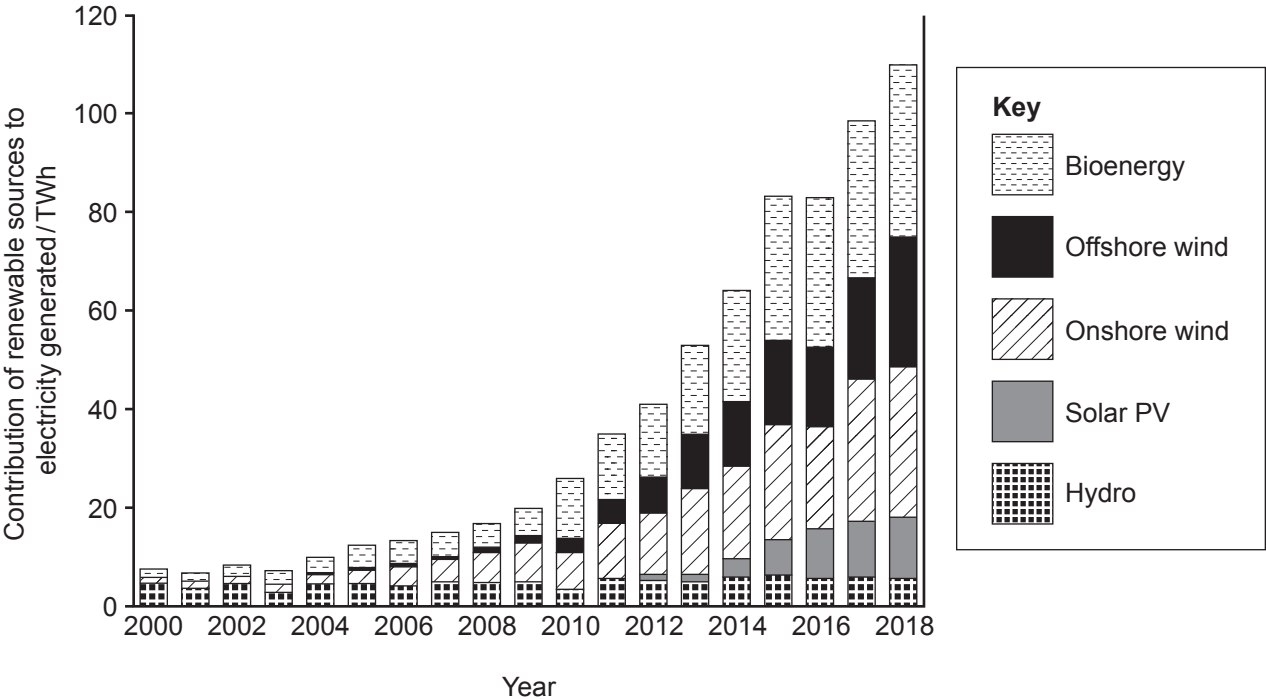


Examiner  
only

(c) The graph shows how renewable sources have contributed to the electricity generated in the UK between 2000 and 2018.



Note: Hydro bar includes shoreline wave/tidal (0.009 TWh in 2018)

(i) Compare the contribution of solar photovoltaic (Solar PV) with hydroelectric power since 2010 **and** suggest a reason for this difference. [2]

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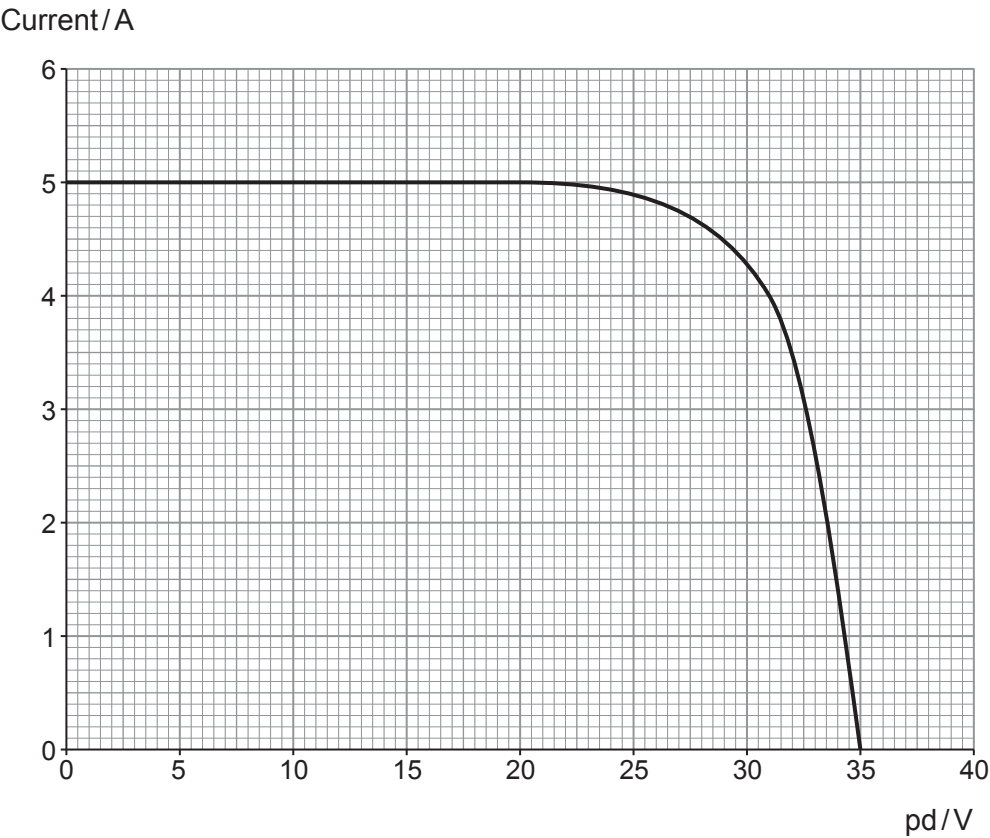
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Examiner  
only

- (ii) A solar panel provides electrical power for a load resistor. The graph below shows the current-potential difference characteristics of a solar panel operating under an intensity of  $600 \text{ W m}^{-2}$ .



The solar panel has a surface area of  $1.2 \text{ m}^2$ . Determine the efficiency of the solar panel when it is operating at its maximum power output. [4]

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- (d) (i) The ground floor of a house has an area of  $60\text{ m}^2$ . The floor is made from concrete of thickness  $120\text{ mm}$ . If the top surface of the concrete is  $19^\circ\text{C}$  and the lower surface is  $5^\circ\text{C}$ , show that the rate of heat flow through the concrete is approximately  $5000\text{ W}$ .  $K_{\text{concrete}} = 0.700\text{ W m}^{-1}\text{ K}^{-1}$  [2]

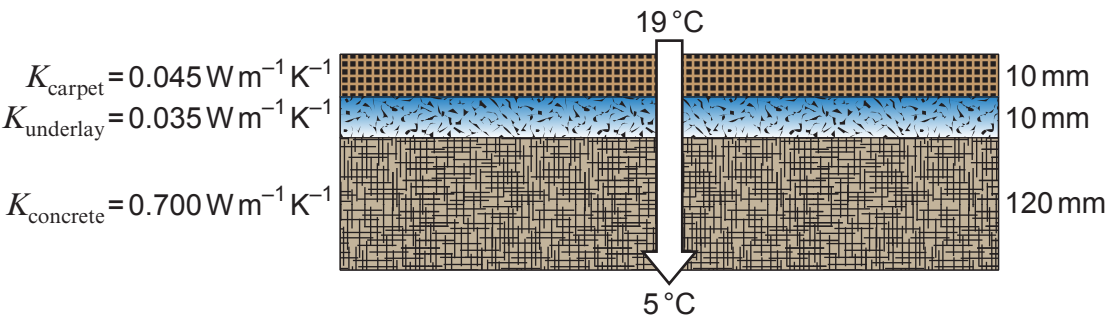
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- (ii) A carpet fitting company adds underlay and carpet on to the concrete as shown in the diagram.

Diagram not drawn to scale



It is stated by the carpet fitting company that the rate of heat flow through the floor will reduce to a quarter of the value calculated in part (d)(i). Evaluate if this claim is correct. [3]

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END OF PAPER

